Choice-Head Distillation for Dental Multiple-Choice Question Answering

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Abstract—Medical large language models (LLMs) have achieved excellent results in benchmark evaluations, but they remain costly to deploy in real-world settings. This challenge is especially pronounced in dental multiple-choice question answering tasks on datasets such as the Chinese Medical Exam dataset (CMExam). Although the output space is restricted to five answer options, many distillation approaches for deployment optimization still train using the full vocabulary distribution. In this paper, we introduce the Choice-Head distillation method, a task-aligned method that transfers only the teacher distribution over the answer options rather than over the entire vocabulary. By distilling in the decision space, our approach is computationally lighter than vocabulary-level distillation and remains compatible with black-box API teachers. In our experiments, DeepSeek-V3 is adopted as the teacher and Qwen2.5-7B and Qwen2.5-14B as students on a CMExam-based resplit. We evaluate the distilled models on a 991-question full test set with repeated runs. The best 14B student reaches 89.10% accuracy, exceeding the 87.18% teacher, while the three-seed mean remains 88.67%. Furthermore, we find that a second hard-label fine-tuning stage is not always beneficial once option-level structure has already been transferred. For structured medical multiple-choice tasks, decision-space (choice-level) distillation is not only cheaper than vocabulary-level distillation, but also yields stronger deployable student models.

Keywords—knowledge distillation; medical question answering; large language models; multiple-choice reasoning; decision-space supervision

I. INTRODUCTION

Knowledge distillation compresses large models by transferring soft targets to smaller students [1], [2], [8]. Recent medical and general-purpose language models also motivate this direction because strong benchmark results often come with high inference cost and limited deployability [4], [5], [9], [10]. In exam-style medical question answering (QA), this trade-off is especially important because the deployment target is often a smaller assistant model rather than the largest teacher model.

Building on the idea of task-specific knowledge distillation, DistilBERT [2] was introduced to perform distillation in the pre-training phase with large-scale unlabeled text. A compressed model using DistilBERT can be further trained for many downstream tasks. Meanwhile, parameter-efficient fine-tuning also provides a practical training mechanism for distillation. LoRA (Low-Rank Adaptation) adapts large models using low-rank incremental parameters, keeping training costs within a manageable range [3]. Combining LoRA with distillation means that the student can receive the teacher's soft-label structural information without incurring the hardware burden of full-parameter updates—an important prerequisite for the rapid iteration of the experimental framework used in this paper.

Stronger teachers and larger evaluation sets have improved reported performance [5], [6], [9]-[13]. In the medical and dental domains, automated clinical chatbots represent one of the most rapidly growing areas of interest within digital medicine and dentistry [14]. To rigorously evaluate their capabilities, a primary assessment strategy involves benchmarking the underlying large language models directly against professional board examination questions. In fixed-choice exams, vocabulary-level targets are often poorly aligned with the downstream decision. Although the underlying models are generative, the downstream task behaves like classification. For a five-option exam, the vast majority of the vocabulary is irrelevant to the final choice. As a result, task-aligned (choice-level) targets simplify training while preserving the uncertainty patterns that determine the correct option. This operational advantage becomes especially important when the teacher is accessed via an external API. In such settings, full-vocabulary distillation is frequently unavailable or too costly because the system cannot provide internal logits at scale. By contrast, option-level targets are easier to collect, easier to verify, and straightforward to align with the evaluation benchmark. Overall, our design reduces not only training complexity, but also data-collection and engineering friction.

In this paper, we study the distillation of dental multiple-choice question answering, which is a simple five-option decision task. Current distillation pipelines use full-vocabulary logits [1], [2] or free-form targets [7]. This output design is highly inefficient for a fixed-choice task with only five options, especially when the original model produces a much larger output space. In addition, cloud-based teachers (API teachers) invoke inference services but do not provide internal logits. Regarding evaluation reliability, existing studies [6], [7] often use small medical QA test sets, which are not fully representative. As a result, model rankings can be unstable: a small number of questions can disproportionately influence the final accuracy. To address this, we adopt a CMExam-based re-splitting strategy, using a 991-question set as the main evaluation set. This makes the test results more robust and more informative for model comparison.

The main contributions of this work are as follows:

1. We propose Choice-Head distillation, which transfers only the teacher distribution over the five answer options. The method is task-aligned for five-option medical MCQs;
2. We show that a student can outperform its teacher: a Qwen2.5-14B student distilled from DeepSeek-V3 reaches 89.10% accuracy on a 991-question CMExam test set, above the teacher accuracy of 87.18%.

II. METHOD

A. *Problem Setting*

While traditional distillation minimizes divergence across the full token vocabulary, our proposed **Choice-Head Distillation** focuses exclusively on the distribution over the relevant option set. Given a multiple-choice question x with five candidate options, the goal is to predict the correct option $y \in \{A, B, C, D, E\}$. Let p_T denote the teacher distribution over these five choices, and let p_S denote the student distribution over the same set. These two targeted distributions are then used to compute the Stage 1 Kullback-Leibler divergence loss (KL loss) below.

B. *Choice-Head Distillation*

The proposed **Choice-Head distillation** reframes the distillation target around the actual decision space of the task. In the five-option multiple-choice setup, the final answer depends on the relative support assigned to A/B/C/D/E rather than on the full token vocabulary. We therefore compare both teacher and student in the same five-dimensional space.

Specifically we isolate the student's logits for the five option tokens at the answer position. Let $z_{S,c}$ be the student logit for option $c \in \{A, B, C, D, E\}$. The student option distribution p_S is obtained by applying a softmax over the five candidate options:

$$p_S(c) = \frac{\exp(z_{S,c})}{\sum_{c' \in \{A,B,C,D,E\}} \exp(z_{S,c'})}$$

The teacher distribution p_T can be obtained from different sources and is used as the teacher term in the Stage 1 KL loss. For a white-box teacher, the option-level log probabilities can be read directly. For an API teacher, the option distribution can be recovered from option-level probabilities when available, or approximated with a smoothed soft label based on the returned answer. In either case, the training signal is restricted to the five answer choices, which makes the method compatible with black-box teachers and cheaper than full-vocabulary distillation.

C. *Stage 1: Choice-Head KL Distillation*

Stage 1 injects the teacher's option-level preference structure into the student. It combines the teacher soft-label distribution with the ground-truth answer provided by the dataset. The loss is defined as:

$$L_{sta} = \alpha \, D_{KL}(p_T \parallel p_S) + (1 - \alpha) L_{CE}.$$

Here, the KL term D_{KL} compares p_T with p_S and encourages the student to match the teacher's relative preference over the five options. The term L_{CE} denotes the cross-entropy loss computed against the gold answer, so it keeps the prediction aligned with the dataset label. The coefficient α controls the balance between teacher-distribution supervision and gold-answer supervision; in the main setting, α is set to 0.35. This stage is intentionally short: for the main 14B student, Stage 1 is trained for one epoch. The goal is structure injection rather than full teacher imitation, because over-training on teacher labels may amplify teacher errors.

D. *Stage 2: Gold-Answer SFT Calibration*

Stage 2 is an optional calibration stage. It inherits the LoRA weights learned in Stage 1 and continues supervised fine-tuning using only the gold answers. Its purpose is to correct possible bias introduced by teacher errors and to pull the student back toward the official answer y^* for input x when the teacher signal is misleading. Here θ denotes the trainable parameters of the student model. The loss is:

$$L_{stage} = -log \, P_\theta(y^* \mid x).$$

Intuitively, Stage 1 gives the student a relative map of the five candidate options, while Stage 2 uses the gold answer y^* to mark the final destination on that map. This second stage can be useful when the student is weak and needs stronger answer-level correction.

III. EXPERIMENTAL SETUP

The evaluation uses a CMExam-based resplit with 6,590 single-choice medical questions across seven subjects [6]. The train, validation, and test splits contain 4,608, 991, and 991 questions. A 125-question dental subset is retained for specialty-focused analysis.

The resplit follows a 70/15/15 partition and is stratified by difficulty, which makes the large test set more representative than the small specialty-only setting used in earlier experiments. A 991-question test does not eliminate uncertainty, but it reduces the chance that a strong result is caused by a favorable sample rather than by a stable improvement in decision quality.

The teacher is DeepSeek-V3 [5]. The students are Qwen2.5-7B-Instruct and Qwen2.5-14B-Instruct [4]. Training uses LoRA with rank 16 and LoRA alpha 32 [3]. For the main 14B setting, Stage 1 is run for one epoch with learning rate 1×10^{-4}.

Teacher labels cover the full training split. In the full-data setting, the teacher and ground truth disagree on about 12.2% of the training questions. This shows that the teacher provides nontrivial relative preferences instead of merely repeating the gold label, while remaining stable enough to serve as a useful transfer source.

Evaluation reports both the 991-question full test set and the 125-question dental subset. The former is the main benchmark used for claims about overall effectiveness. The latter is retained for domain-specific inspection, but its smaller size makes it more sensitive to variance and therefore less suitable as the sole basis for a central conclusion.

To reduce run-to-run noise, the main comparisons use repeated runs under the same data split and prompt format. This matters because a student-over-teacher claim is only meaningful if the margin is reproducible across seeds rather than produced by a single favorable trial. The reported mean and best results are therefore both useful: the mean reflects stability, while the best run shows the upper bound of the same training recipe.

All evaluations use exact-match accuracy on the final option choice. The prompt format is kept fixed across teacher and student runs so that the comparison reflects model quality rather than prompt engineering differences. This protocol is intentionally simple. The goal of the paper is not to maximize performance through elaborate prompting, but to isolate the effect of decision-space supervision under a controlled and reproducible setup.

The evaluation protocol also keeps the answer space strictly closed during both label generation and testing. Each question is decoded into one of the five predefined options rather than into free-form natural language. This detail matters because it removes a common source of noise in medical QA benchmarking: a model may express a clinically reasonable explanation while still failing to commit to a valid option string. By constraining teacher and student outputs to the same closed decision interface, the reported accuracy measures the quality of the final answer choice itself. That makes the comparison cleaner and better aligned with the conference paper's central claim, which concerns decision-space transfer rather than general text generation quality.

IV. RESULTS AND DISCUSSION

Main Findings and Evaluation Strategy: Our primary evaluation is conducted on the full 991-question test set rather than the 125-question dental subset. While the dental subset remains valuable for verifying whether performance gains generalize to specific target specialties, its restricted size makes it highly sensitive to minor, question-level variances. Consequently, we treat the dental subset as supporting evidence rather than the primary metric for model benchmarking and ranking.

Choice-Head distillation improves both student sizes. On the full 991-question test set, the 7B student rises from 76.49% to 85.60%, a gain of 9.11 percentage points. The 14B student rises from 83.55% to 88.67% on average, a gain of 5.12 percentage points. The best 14B run reaches 89.10%, exceeding the 87.18% teacher.

The result is also stable. Across three seeds, the 14B distilled model ranges from 88.40% to 89.10%, a spread of only 0.70 percentage points. The mean result and the best run therefore support the same conclusion: ***the task-aligned student is consistently competitive with, and in the best case stronger than, the teacher***.

Table 1. Results on the CMExam full-data and dental test sets. (Full Test = 991-question full test set. Dental Test = 125-question dental subset.)

Model	Setting	Full Test Accuracy	Dental Test Accuracy
Qwen2.5-7B	Zero-shot baseline	76.49%	68.80%
Qwen2.5-14B	Zero-shot baseline	83.55%	74.40%
DeepSeek-V3	Teacher	87.18%	79.20%
Qwen2.5-7B	Stage 1 mean	85.60%	73.60%
Qwen2.5-14B	Stage 1 mean	88.67%	79.20%
Qwen2.5-14B	Stage 1 best	89.10%	78.40%

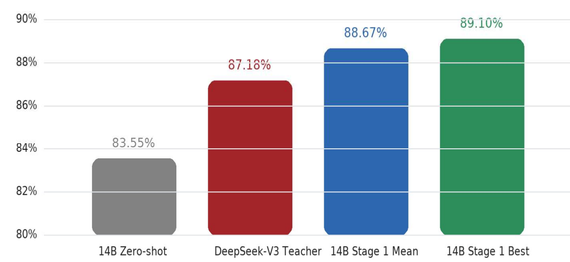

Figure 1. Main results on the 991-question full test set. The distilled 14B student improves over the zero-shot baseline and exceeds the teacher; the best run reaches 89.10%, while the three-seed mean remains 88.67%.

The Role of Stage 2 SFT Calibration: Additional Stage 2 runs show that supplementary gold-answer fine-tuning does not consistently improve performance after Stage 1. For both 7B and 14B students, the best or average results do not depend on an additional Stage 2 update, suggesting that hard-label calibration can weaken useful soft-label structure after the option-level signal has already been transferred. Therefore, Stage 2 should be treated as a conditional calibration step rather than as a fixed part of a longer-is-better pipeline; its usefulness depends on student capacity, teacher reliability, and data scale.

Student-Over-Teacher Dynamics: The phenomenon of the student outperforming the teacher does not imply that the teacher model lacks general capability. Instead, a more plausible explanation is that the student is optimized within a narrower, more precisely aligned decision space. While the teacher provides rich, relative preferences across the five options (A/B/C/D/E), the training dataset enforces the factual constraints of the gold-standard answers. Synthesizing these dual signals within a restricted five-option output space yields a smoother, more task-specific decision boundary than training on hard labels alone.

Scope of Applicability: Overall, these results strongly validate the central thesis of this work. For structured, five-option medical examinations, knowledge distillation should directly target the exact decision space dictated by the evaluation metric. This targeted approach concentrates the supervision signal, eliminates irrelevant vocabulary-level noise, and yields compact student models that are significantly more viable for real-world deployment. Importantly, these conclusions should not be broadly generalized to open-ended clinical dialogue or rationale generation. Tasks requiring complex, generative language behaviors inevitably demand broader vocabulary supervision and distinct alignment objectives.

V. Practical Implications and Limitations

The main practical implication is that decision-space distillation changes the deployment trade-off. Instead of asking whether a smaller student can perfectly reproduce a large teacher, the method asks whether the student can preserve the part of the teacher signal that directly affects the final option choice. For standardized medical MCQs, that is the decision that matters in practice. This framing is useful for educational systems, exam preparation tools, and low-cost assessment services, where inference cost, model size, and operational simplicity matter as much as benchmark accuracy.

The results also clarify when the method should be preferred over more general distillation objectives. If the downstream task is evaluated by a small and explicit answer set, then transferring a full-vocabulary distribution may add cost without adding useful supervision. In that setting, option-level transfer provides a cleaner training signal and a simpler debugging path. When the student fails, the failure can be inspected directly at the answer-choice level instead of being buried in a large token distribution that is only indirectly related to the final decision.

At the same time, the method has clear limits. The present experiments focus on fixed-choice dental and medical exam questions. They do not show that the same target is sufficient for open-ended diagnosis, rationale generation, or interactive clinical dialogue. Those tasks depend on broader language behavior and would require a different supervision target. The current evidence therefore supports a task-specific claim: for structured five-option exams, decision-space supervision is effective; it should not yet be treated as a general replacement for language-model distillation.

Another limitation is that the current study evaluates a single strong teacher and a narrow family of student architectures. That is enough to establish the central result, but not enough to show that every teacher-student pair will behave the same way. Future work should test whether the same pattern holds across more teacher qualities, more student sizes, and more medical subdomains. A second extension is to combine decision-space supervision with selective data filtering, so that ambiguous or low-value training cases are handled differently from clear high-signal examples.

VI. Conclusion

This paper presented Choice-Head distillation for dental multiple-choice question answering. The method distills only the five-option answer distribution, not the full vocabulary. This makes the supervision target closer to the task and keeps the framework compatible with black-box teachers. On the 991-question CMExam-based test set, the best 14B student reaches 89.10% accuracy and exceeds the 87.18% DeepSeek-V3 teacher. The corresponding three-seed mean of 88.67% shows that the gain is not restricted to a single run. For structured medical multiple-choice tasks, decision-space distillation is therefore a practical path to smaller and more deployable QA systems.

References

[1] G. Hinton, O. Vinyals, and J. Dean, "Distilling the Knowledge in a Neural Network," in NIPS Deep Learning and Representation Learning Workshop, Montreal, Canada, 2015. Available: https://arxiv.org/abs/1503.02531

[2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," arXiv:1910.01108, 2019. Available: https://arxiv.org/abs/1910.01108

[3] E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," arXiv:2106.09685, 2021. Available: https://arxiv.org/abs/2106.09685

[4] Qwen Team, "Qwen2.5 Technical Report," arXiv:2412.15115, 2024. Available: https://arxiv.org/abs/2412.15115

[5] DeepSeek-AI, "DeepSeek-V3 Technical Report," arXiv:2412.19437, 2024. Available: https://arxiv.org/abs/2412.19437

[6] T. Liu et al., "Benchmarking Large Language Models on CMExam: A Comprehensive Chinese Medical Exam Dataset," in Advances in Neural Information Processing Systems, 2023, doi: 10.52202/075280-2283.

[7] H. Zhang et al., "HuatuoGPT, Towards Taming Language Models to Be a Doctor," in Findings of the Association for Computational Linguistics: EMNLP 2023, 2023. [Online]. Available: https://aclanthology.org/2023.findings-emnlp.725/

[8] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," International Journal of Computer Vision, vol. 129, no. 6, pp. 1789-1819, 2021, doi: 10.1007/s11263-021-01453-z.

[9] K. Singhal et al., "Large language models encode clinical knowledge," Nature, vol. 620, pp. 172-180, 2023, doi: 10.1038/s41586-023-06291-2.

[10] A. J. Thirunavukarasu et al., "Large language models in medicine," Nature Medicine, vol. 29, pp. 1930-1940, 2023, doi: 10.1038/s41591-023-02448-8.

[11] R. C. W. Chau, K. M. Thu, O. Y. Yu, T.-C. Hsung, E. C. M. Lo, and W. Y. H. Lam, "Performance of Generative Artificial Intelligence in Dental Licensing Examinations," International Dental Journal, vol. 74, no. 3, pp. 616–621, Jun. 2024, doi: 10.1016/j.identj.2023.12.007.

[12] R. C. W. Chau, K. M. Thu, O. Y. Yu, E. C. M. Lo, T.-C. Hsung, and W. Y. H. Lam, "Response to Generative AI in Dental Licensing Examinations: Comment," International Dental Journal, vol. 74, no. 4, pp. 897–898, Aug. 2024, doi: 10.1016/j.identj.2024.02.002.

[13] R. C. W. Chau et al., "Evaluation of Chatbot Responses to Text-Based Multiple-Choice Questions in Prosthodontic and Restorative Dentistry," Dentistry Journal, vol. 13, no. 7, p. 279, Jul. 2025, doi: 10.3390/dj13070279.

[14] W. Y. H. Lam et al., "Digital Dentistry in Clinical Practice: A Scoping Review of Current Capabilities and Future Directions," International Dental Journal, vol. 76, no. 1, Feb. 2026, Art. no. 109296, doi: 10.1016/j.identj.2025.109296.